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Sample Name # MINERALOGRAMA EN CABELLO
DEMO

Requested by # Teletest Analytika SA
Sample Number **18267691** Receipt Data: 02/10/2026 11:10

Report Validation and Issue Date **02/10/2026 11:28** Centro Demostración

Hair Mineral Composition



MINERALOGRAMA EN CABELLO

Date **02-10-2026**

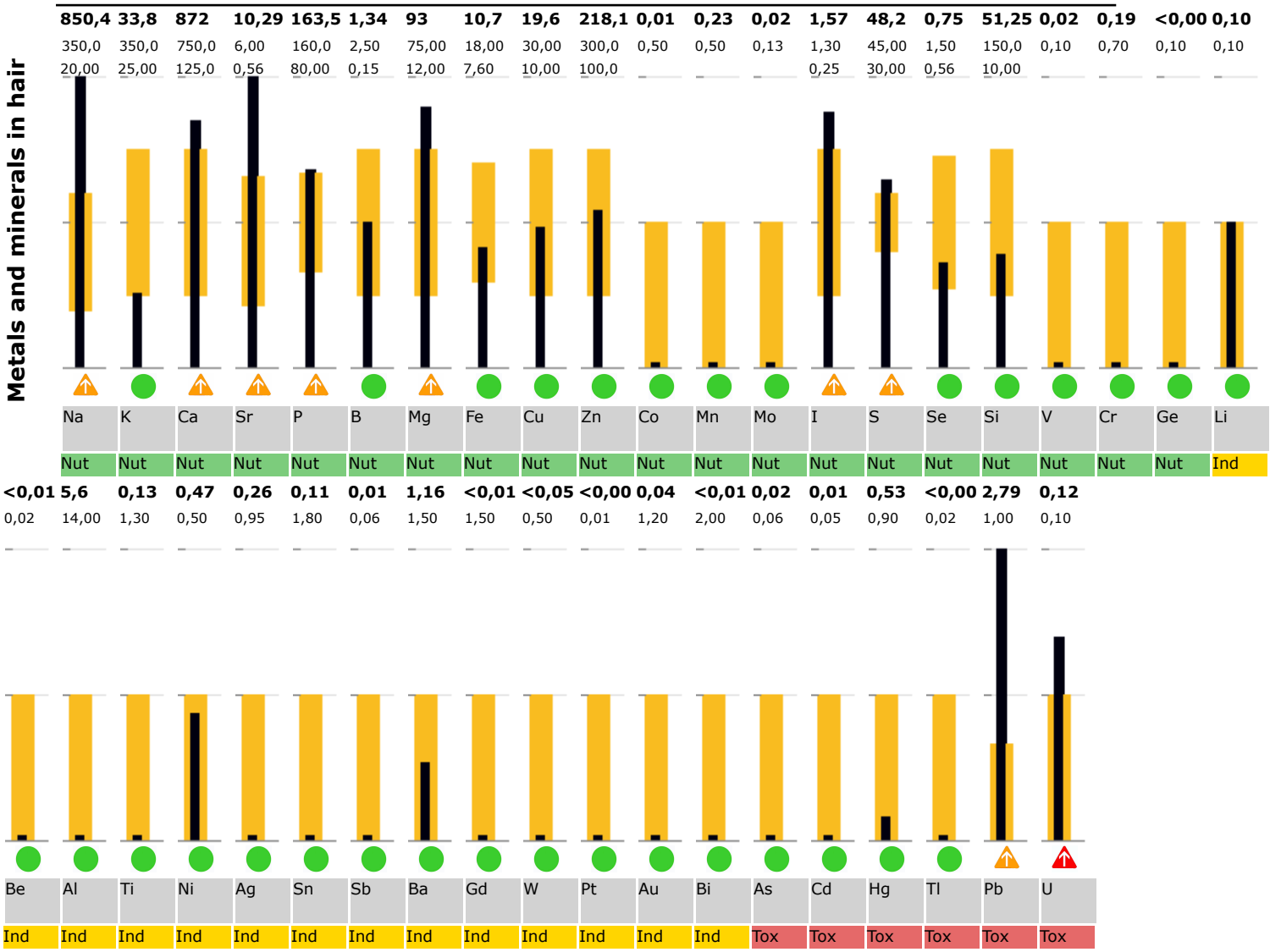
Requested by # **Teletest Analytika SA**

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SEGURIDAD ALIMENTARIA Y AMBIENTAL (AGUAS, ALIMENTOS, SUPERFICIES, AMBIENTES). TÉCNICAS INSTRUMENTALES (AGUAS, ALIMENTOS, AMBIENTES, MUESTRAS BIOLÓGICAS). HIGIENE INDUSTRIAL, AMIANTO EN FILTROS Y MATERIALES.
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Test Report
Report **18267691**v1
Customer **DEMOSTRACION**
Customer Reference # DEMO

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Hair Mineral Composition



850,4	350,0	20,00	
33,8	350,0	25,00	
872	750,0	125,0	
10,29	6,00	0,56	
163,5	160,0	80,00	
1,34	2,50	0,15	
93	75,00	12,00	
10,7	18,00	7,60	
19,6	30,00	10,00	
218,1	300,0	100,0	
0,01	0,50		
0,23	0,50		
0,02	0,13		
1,57	1,30	0,25	
48,2	45,00	30,00	
0,75	1,50	0,56	
51,25	150,0	10,00	
0,02	0,10		
0,19	0,70		
<0,00	0,10		
0,10	0,10		

<0,01	0,02		
5,6	14,00		
0,13	1,30		
0,47	0,50		
0,26	0,95		
0,11	1,80		
0,01	0,06		
1,16	1,50		
<0,01	1,50		
<0,05	0,50		
<0,00	0,01		
0,04	1,20		
<0,01	2,00		
0,02	0,06		
0,01	0,05		
0,53	0,90		
<0,00	0,02		
2,79	1,00		
0,12	0,10		

Indices

Ca / Mg	Ratio 9	*(RV:3/18Ratio)*
Ca / K	Ratio **25,8**	*(RV:0,7/12,0Ratio)*
Ca / P	Ratio **5,3**	*(RV:1,0/5,0Ratio)*
Ca / Na	Ratio **1,0**	*(RV:1,5/14,0Ratio)*
Ca / Fe	Ratio **81**	*(RV:11/45Ratio)*
Ca / Zn	Ratio 4,0	*(RV:1,0/5,0Ratio)*
Na / K	Ratio **25,2**	*(RV:0,1/4,0Ratio)*
Na / Mg	Ratio **9,1**	*(RV:0,6/6,0Ratio)*
Zn / Cu	Ratio 11,1	*(RV:4,0/16,0Ratio)*
Fe / Cu	Ratio 0,5	*(RV:0,4/1,8Ratio)*

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Metals and minerals in hair

The metals and minerals in hair are used as an indicator of the accumulation in our tissues of both minerals and toxins.
They are also related to neurological pathologies
They are analyzed by ICP-MS/MS.

Sodium [Na] **850,4** µg/g hair *(RV:20,0/350,0µg/g hair)*
Nutritious



Sodium (Na) is an essential mineral and the main extracellular cation, vital for maintaining fluid balance, osmotic pressure, and blood volume. It participates in the transmission of nerve impulses and muscle contraction, regulating, along with potassium, the cell membrane potential. It is obtained primarily through table salt, processed foods, sausages, and canned goods.

Potassium [K] 33,8 µg/g hair *(RV:25,0/350,0µg/g hair)*
Nutritious



Potassium (K) is the main intracellular cation, essential for maintaining acid-base balance, muscle contraction and nervous transmission. It participates in protein synthesis and cell enzyme function. The main food sources are fruit (banana, avocado), vegetables, legumes and nuts.

Calcium [Ca] **872** µg/g hair *(RV:125/750µg/g hair)*
Nutritious



Calcium (Ca) is the most abundant mineral in the body, present mainly in bones and teeth, where it provides structure and resistance. It participates in muscle contraction, blood clotting, nervous transmission and enzyme activation. The main sources are dairy products, fish with thorns, green-leaf vegetables and nuts.

Strontium [Sr] **10,29** µg/g hair *(RV:0,56/6,00µg/g hair)*
Nutritious



Strontium (Sr) es un metal alcalinotérreo que se encuentra en bajas concentraciones en el cuerpo, depositándose principalmente en huesos, donde puede sustituir parcialmente al calcio, favoreciendo la formación de tejido óseo y disminuyendo la resorción ósea, por lo que se estudia en el tratamiento de osteoporosis. Se encuentra en el agua potable, vegetales y mariscos.

Phosphorus [P] **163,5** µg/g hair *(RV:80,0/160,0µg/g hair)*
Nutritious



Phosphorus (P) is a macromineral present in the form of phosphate, an essential component of ATP, nucleic acids, phospholipids, and bones, playing a key role in energy metabolism and the regulation of acid-base balance. The main sources are meat, fish, dairy products, legumes, and nuts.

Boron [B] 1,34 µg/g hair *(RV:0,15/2,50µg/g hair)*
Nutritious



Boron (B) is an oligoelement involved in the metabolism of minerals such as calcium, magnesium and phosphorus, and in the regulation of steroid hormones and vitamin D, promoting bone health and cognitive function. It has anti-inflammatory properties and can modulate antioxidant enzymes. It is found in fruits (apples, pears), green-leaf vegetables, nuts and legumes.

Magnesium [Mg] **93** µg/g hair *(RV:12/75µg/g hair)*
Nutritious



Magnesium (Mg) is a macromineral that acts as a cofactor in more than 300 enzyme reactions, participating in protein synthesis, energy production, muscle contraction, nervous function and bone health. It is involved in the regulation of heart rate and control of blood glucose. It is found in green-leaf vegetables, nuts, seeds, legumes and whole cereals.

Iron [Fe] 10,7 µg/g hair *(RV:7,6/18,0µg/g hair)*
Nutritious



Iron (Fe) is an essential trace mineral involved in the formation of hemoglobin and myoglobin, facilitating the transport and storage of oxygen in tissues. It plays a role in immune function, DNA synthesis, and oxidative reactions. It is found in red meat, liver, legumes, spinach, and whole grains.

Copper [Cu] 19,6 µg/g hair *(RV:10,0/30,0µg/g hair)*
Nutritious



Copper (Cu) is an essential oligoelement that is part of antioxidant enzymes (superoxide dismutase), involved in the formation of collagen, iron metabolism and immune function. It is found in shellfish, liver, nuts and seeds.

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Zinc [Zn]	218,1 µg/g hair	*(RV:100,0/300,0µg/g hair)*	
Nutritious			

Zinc (Zn) es un oligoelemento esencial que actúa en más de 300 enzimas relacionadas con la síntesis de proteínas, la función inmune, la cicatrización, la salud reproductiva y la antioxidación. Se encuentra en carnes, mariscos, huevos, legumbres y semillas.

Cobalt [Co]	0,01 µg/g hair	*(RV: ≤0,50µg/g hair)*	
Nutritious			

Cobalt (Co) forms part of the structure of vitamin B12, which is essential for the formation of red blood cells and the maintenance of neurological health. The sources are foods of animal origin such as meat, fish, eggs and dairy.

Manganese [Mn]	0,23 µg/g hair	*(RV: ≤0,50µg/g hair)*	
Nutritious			

Manganese (Mn) is an oligoelement involved in the metabolism of carbohydrates, amino acids and lipids, bone formation and as a cofactor of antioxidant enzymes. It is found in whole cereals, nuts, tea and green-leaf vegetables.

Molybdenum [Mo]	0,02 µg/g hair	*(RV: ≤0,13µg/g hair)*	
Nutritious			

Molybdenum (Mo) is a oligoelement that acts as a cofactor in enzymes related to the metabolism of sulphur and purines (xantine oxidase, sulfite oxidase), participating in detoxification processes. It is found in vegetables, cereales and nuts.

Iodine [I]	**1,57** µg/g hair	*(RV:0,25/1,30µg/g hair)*	
Nutritious			

Iodine (I) is essential for the synthesis of thyroid hormones (T3 and T4), which regulate metabolism, growth, and development. It is found in iodized salt, fish, shellfish, and seaweed.

Sulfur [S]	**48,2** mg/ g hair	*(RV:30,0/45,0mg/ g hair)*	
Nutritious			

Sulfur (S) is a component of essential amino acids (cysteine, methionine) and compounds such as glutathione, participating in hepatic detoxification and collagen synthesis. It is found in protein foods of animal and plant origin.

Selenium [Se]	0,75 µg/g hair	*(RV:0,56/1,50µg/g hair)*	
Nutritious			

Selenium (Se) es un oligoelemento con funciones antioxidantes, formando parte de la glutatión peroxidasa y participando en el metabolismo de hormonas tiroideas. Se encuentra en nueces de Brasil, pescado, mariscos y cereales.

Silicon [Si]	51,25 µg/g hair	*(RV:10,00/150,00µg/g hair)*	
Nutritious			

Silicon (Si) is a mineral involved in collagen synthesis and the health of bones, skin, hair, and nails. It plays a role in bone mineralization and the integrity of connective tissue. It is found in whole grains, bananas, and vegetables.

Vanadium [V]	0,02 µg/g hair	*(RV: ≤0,10µg/g hair)*	
Nutritious			

Vanadium (V) is an oligoelement with a possible role in the metabolism of lipids and glucose, partially imitating the action of insulin. It is found in seafood, mushrooms and cereals.

Chromium [Cr]	0,19 µg/g hair	*(RV: ≤0,70µg/g hair)*	
Nutritious			

Chromium (Cr) is an oligoelement that potentiates the action of insulin, improving the metabolism of carbohydrates, proteins and fats. It is found in meat, whole cereals, broccoli and beer yeast.

Germanium [Ge]	<0,002 µg/g hair	*(RV: ≤0,10µg/g hair)*	
Nutritious			

Germanium (Ge) es un oligoelemento con potencial actividad antioxidante e inmune, pero no se considera esencial. Se encuentra en ajo, ginseng y setas.

Lithium [Li]	0,10 µg/g hair	*(RV: ≤0,10µg/g hair)*	
Indicators			

Lithium (Li) is an alkaline metal used as a mood stabilising drug in bipolar disorders, regulating neurotransmitters and intracellular signalling. The sources of exposure are therapeutic use, drinking water in areas with lithium and occupational exposure in battery and ceramic industries. Their accumulation causes toxicity with tremors, ataxia, diarrhoea, vomiting, severe kidney and neurological disorders.

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Beryllium [Be] <0,01 µg/g hair *(RV: ≤0,02µg/g hair)*
Indicators

Beryllium (Be) is an alkalinotereal metal without biological function, used in alloys, aerospace and electronic industry. It is mainly exposed by inhalation in industrial environments. Its accumulation produces beriliosis (chronic granulomatous pneumonitis), pulmonary hypersensitivity and lung cancer.

Aluminum [Al] 5,6 µg/g hair *(RV: ≤14,0µg/g hair)*
Indicators

Aluminium (Al) is an abundant metal in the earth's crust without essential biological function. The sources of exposure are: kitchen utensils, food additives, antacid medicines, vaccines and environmental pollution. It accumulates in bone, lung, and brain. Their accumulation is related to neurotoxicity, encephalopathy, bone disorders and possible relationship with neurodegenerative diseases such as Alzheimer's in chronic exposures.

Titanium [Ti] 0,13 µg/g hair *(RV: ≤1,30µg/g hair)*
Indicators

Titanium (Ti) is a metal that has no known biological functionality, used in prostheses, pigments (titanium dioxide) and cosmetics. Exposure occurs from medical implants or industrial dust inhalation. It is not considered toxic at normal levels, but titanium dioxide can cause pulmonary inflammation by inhaling at high concentrations

Nickel [Ni] 0,47 µg/g hair *(RV: ≤0,50µg/g hair)*
Indicators

Nickel (Ni) is a non-essential metal, although in some organisms it participates in enzymes. It is used in alloys, jewelry, batteries and coin coatings. Exposure occurs through dermal and occupational contact. Its accumulation produces contact dermatitis, awareness, lung toxicity, and risk of nasal and lung cancer in chronic industrial exposures.

Silver [Ag] 0,26 µg/g hair *(RV: ≤0,95µg/g hair)*
Indicators

Silver (Ag) is a metal used in jewelry, utensils, photography, and medicine (as an antimicrobial in dressings). Exposure occurs through contact, ingestion of colloidal silver supplements, and occupational exposure. Accumulation can cause argyria (a bluish-gray discoloration of the skin and mucous membranes), and liver and kidney toxicity at high exposure levels.

Tin [Sn] 0,11 µg/g hair *(RV: ≤1,80µg/g hair)*
Indicators

Tin (Sn) is a metal used in cans, alloys, and plastics. Exposure comes from canned foods and industrial settings. It is generally not very toxic in its inorganic form, but organotin compounds can cause neurotoxicity, gastrointestinal disturbances, and liver damage.

Antimony [Sb] 0,01 µg/g hair *(RV: ≤0,06µg/g hair)*
Indicators

Antimony (Sb) is a metal used in flame retardants, batteries and alloys. It is mainly exposed by inhalation in industrial environments. Their accumulation may cause respiratory irritation, gastrointestinal disorders, hepatotoxicity and possible cancer effects in chronic exposures.

Barium [Ba] 1,16 µg/g hair *(RV: ≤1,50µg/g hair)*
Indicators

Bario (Ba) is a metal used in industry, explosives and radiological studies as barium sulphate (non-toxic). Soluble compounds are toxic, and exposure occurs by inhalation or ingestion of soluble salts. Its toxicity causes hypokalaemia, muscle palsy, arrhythmias and respiratory disorders.

Gadolinium [Gd] <0,01 µg/g hair *(RV: ≤1,50µg/g hair)*
Indicators

Gadolinium (Gd) is a metal used in contrast agents for magnetic resonance. Exposure occurs by injection during medical studies. Their accumulation may cause nephrogenic systemic fibrosis in patients with renal impairment and brain tissue deposits at repeated exposures.

Tungsten [W] <0,05 µg/g hair *(RV: ≤0,50µg/g hair)*
Indicators

Tungsten (W) or Wolframian is a metal used in light-bulb filaments, tools and alloys. Exposure occurs in industrial environments. It usually has low toxicity, but some compounds can cause pulmonary irritation and possible kidney disorders.

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Platinum [Pt]	<0,001 µg/g hair	*(RV: ≤0,010µg/g hair)*	
Indicators			

Platinum (Pt) is a metal used in jewelry, catalysts and chemotherapy (cisplatin). Exposure occurs in industries and medical treatments. It may cause hypersensitivity, nephrotoxicity, ototoxicity and neurotoxicity in the case of its pharmacological compounds.

Gold [Au]	0,04 µg/g hair	*(RV: ≤1,20µg/g hair)*	
Indicators			

Gold (Au) is a metal used in jewelry, dentistry, and treatments for rheumatoid arthritis (gold salts). Exposure occurs through medical use or contact with objects. Gold salts can cause dermatitis, membranous nephropathy, thrombocytopenia, and allergic reactions.

Bismuto [Bi]	<0,01 µg/g hair	*(RV: ≤2,00µg/g hair)*	
Indicators			

Bismuth (Bi) is a metal used in medicines (bismuth subsalicylate) and cosmetics. The main exposure is medical. In excess it can cause neurotoxicity, encephalopathy and renal damage, although at therapeutic doses it is safe.

Arsenic [As]	0,02 µg/g hair	*(RV: ≤0,06µg/g hair)*	
Toxic			

Arsenic (As) is a metal found in contaminated waters, pesticides and some foods. Chronic exposure causes skin, lung and bladder cancer, peripheral neuropathies, skin and cardiovascular disorders. It is one of the most toxic and priority metals in public health.

Cadmium [Cd]	0,01 µg/g hair	*(RV: ≤0,05µg/g hair)*	
Toxic			

Cadmium (Cd) is a metal found in batteries, welds and tobacco. Exposure occurs by inhalation, consumption of contaminated food or tobacco. Their accumulation causes renal damage, bone demineralization (Itai-Itai) and lung cancer.

Mercury [Hg]	0,53 µg/g hair	*(RV: ≤0,90µg/g hair)*	
Toxic			

Mercury (Hg) is a highly toxic metal in all its forms (elemental, inorganic, methylmercury). The sources of exposure are: dental amalgams, contaminated fish, industry and artisanal mining. Its toxicity causes neurotoxicity, kidney disorders, tremors, cognitive impairment and immune disorders.

Thallium [Tl]	<0,001 µg/g hair	*(RV: ≤0,020µg/g hair)*	
Toxic			

Thallium (Tl) is a metal used in electronics and pesticides. Exposure occurs by environmental or industrial pollution. It is highly toxic, causing peripheral neuropathies, alopecia, gastrointestinal, heart and renal damage.

Lead [Pb]	**2,79** µg/g hair	*(RV: ≤1,00µg/g hair)*	
Toxic			

Lead (Pb) is a metal used historically in paints, pipes and gasoline. Exposure occurs from dust, contaminated water and industrial environments. Their accumulation causes neurotoxicity (especially in children), anaemia, hypertension, kidney disorders and reproductive disorders.

Uranium [U]	**0,12** µg/g hair	*(RV: ≤0,10µg/g hair)*	
Toxic			

Uranium (U) is a metal that is used as nuclear fuel but also found in its natural form on certain lands. Exposure occurs in mining, contaminated water or inhalation. It is toxic for its radiation (cancer risk) and chemical toxicity (nephrotoxicity).

Hair Mineral Composition

*(**RV:** Reference values expressed in italics)*

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Report Validated by **Carolina Garcia Sabaté**
Clinical Analyst. CBC 22319-C
Report Validation and Issue Date 02-10-2026 11:28

Report issued by: **TeleTest Analytika S.A.**. Tel 932 123 345; laboratory@teletest.es

18267691 **tFjbW**

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Clinical Analysis

Generalitat de Catalunya Authorization Health Department E08750211. Provenza 169 Local 2. 08036, Barcelona. Spain. CIF ES A58023326.
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Legend Symbols



Low .... Reference Value .... High

Interpret according to patient data

Result with comments

Indeterminate or Inconclusive

Result
VR Max
VR Min

Antecedent, if applicable



Reference Value

Result

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